HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

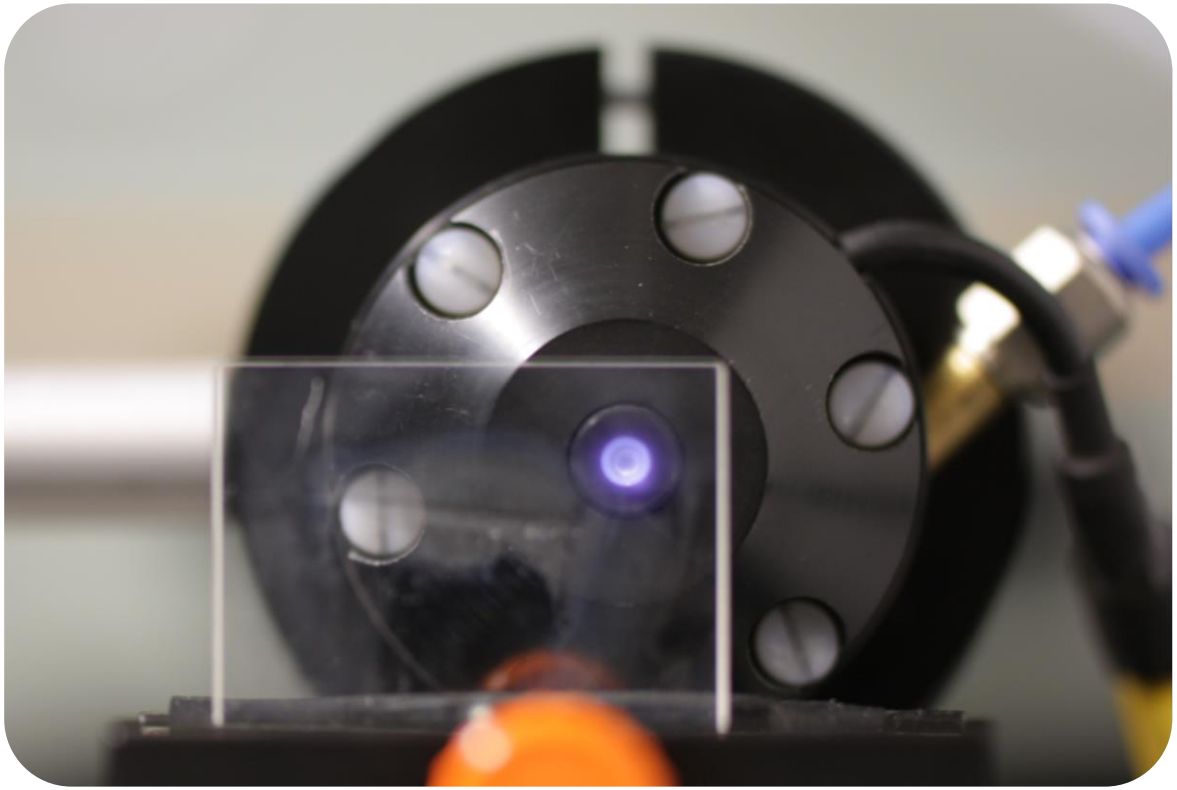
Exercises and Worked Examples + Solutions

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'
Chapter 3

Prof. Dr. Christoph Gerhard, 2024

3 Plasmas in materials science
(and processing)



Plasma-chemical mechanisms

Which ones are the 'chemical' mechanisms?

- **Selection of plasma-surface interactions**
  → Plasma surface processing mechanisms
  - Plasma-induced oxidation/reduction
    - → Cleaning, hardening, modification…
  - Plasma heating
    - → Thermal surface modification
  - Plasma-induced formation/orientation of dipoles
    - → Electrostatic surface activation
  - UV plasma emission
    - → Cleaning
  - Plasma polymerisation
    - → Coating
  - Plasma-induced attachment of functional groups
    - → Chemical surface activation
  - Ion bombardment
    - → Material removal (sputtering), plasma etching, cleaning
  - etc.

Plasma-chemical mechanisms

- **Selection of (chemical) plasma-surface interactions**
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  - etc.

Plasma-physical mechanisms

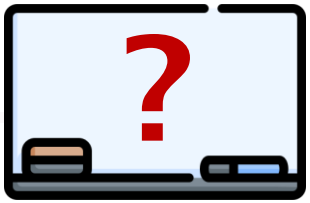
Which ones are the 'physical' mechanisms?

- **Selection of plasma-surface interactions**
  → Plasma surface processing mechanisms
  - Plasma-induced oxidation/reduction
    - → Cleaning, hardening, modification…
  - Plasma heating
    - → Thermal surface modification
  - Plasma-induced formation/orientation of dipoles
    - → Electrostatic surface activation
  - UV plasma emission
    - → Cleaning
  - Plasma polymerisation
    - → Coating
  - Plasma-induced attachment of functional groups
    - → Chemical surface activation
  - Ion bombardment
    - → Material removal (sputtering), plasma etching, cleaning
  - etc.

Plasma-physical mechanisms

- **Selection of (physical) plasma-surface interactions**
  → Plasma surface processing mechanisms
  - Plasma-induced oxidation/reduction
    - → Cleaning, hardening, modification…
  - Plasma heating
    - → Thermal surface modification
  - Plasma-induced formation/orientation of dipoles
    - → Electrostatic surface activation
  - UV plasma emission
    - → Cleaning
  - Plasma polymerisation
    - → Coating
  - Plasma-induced attachment of functional groups
    - → Chemical surface activation
  - Ion bombardment
    - → Material removal (sputtering), plasma etching, cleaning
  - etc.

Precision cleaning



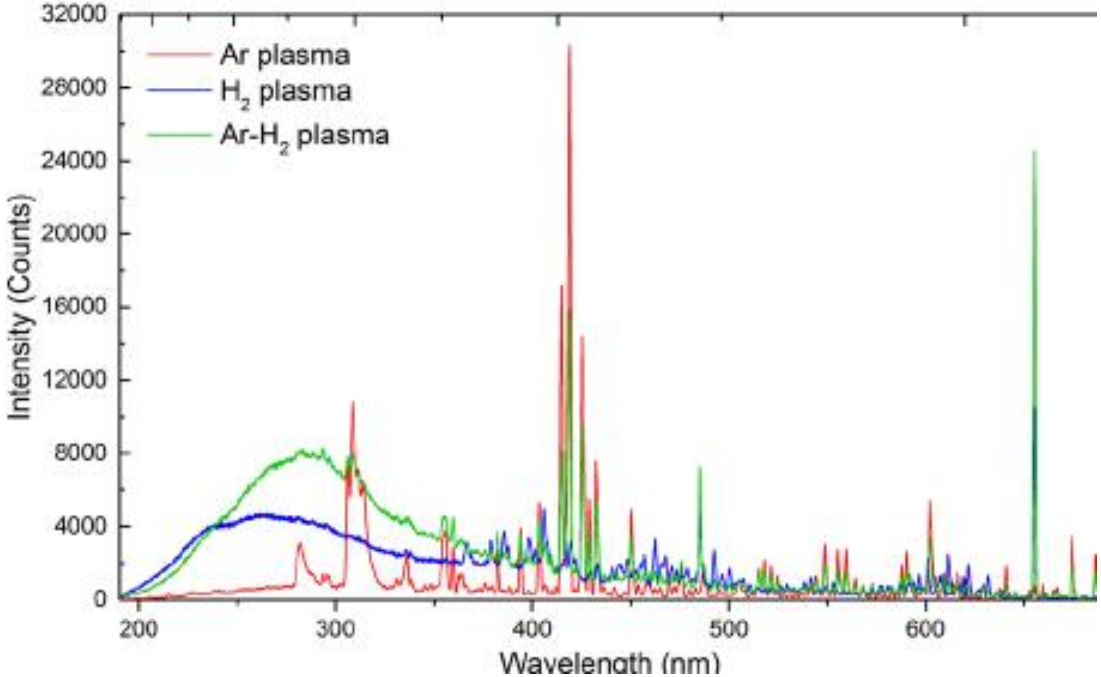
- **Worked example 3.1**
  - The task is to remove a silicon monoxide layer from a wafer made of silicon applying a plasma. Which plasma-induced effects or mechanisms could be used/could contribute to such cleaning?

Precision cleaning

- **Worked example 3.1 – Answer**
  - Possible effects/mechanisms for cleaning:
    - Removal of oxygen
      - via reduction:
        $SiO(s) + N(p)$ or $H(p) \rightarrow Si(s) + H_2O(g)$ or $NO_x\ (g)$
      - via kinetic energy of incident ions due to sheath-effect
      - via kinetic energy of electrons accelerated by the electric field
      - via photons (plasma UV emission)
      - via thermal decomposition and vaporisation (plasma heating)
      - tbc…

Precision cleaning

- **Worked example 3.2**
  - The task is to remove a SiO oxide layer from a silicon wafer by indirect plasma treatment. The available process gases are argon, hydrogen, and Varigon (i.e., a mixture of argon and hydrogen), leading to the plasma emission characteristics shown below. Which process gas should be used in this case? Please give a quantitative answer. Hint: The binding energy of SiO is 451 kJ/mol.



DOI:10.1063/1.4941298

Precision cleaning

- **Worked example 3.2 – Answer**

$$\Delta H_{Si-O} = 451\ kJ/mol$$

$$E = \frac{451000\ J}{6.022 \cdot 10^{23}} = \underline{7.49 \cdot 10^{-19}\ J}$$

Definition : $E = E_{photon}$

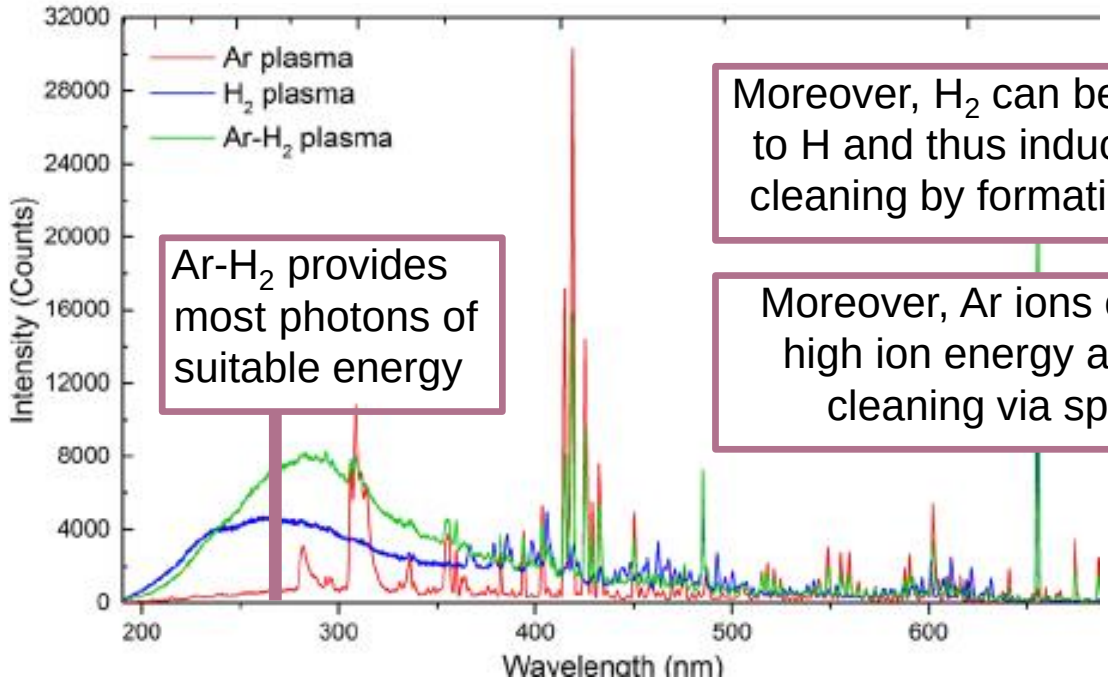
$$E_{photon} = h \cdot f \rightarrow f = \frac{7.49 \cdot 10^{-19}\ J}{6.626 \cdot 10^{-34}\ Js}$$

$$= \underline{1.13 \cdot 10^{15}\ Hz}$$

Precision cleaning

- **Worked example 3.2 – Answer**
  - Wavelength ('photon energy') needed for decomposition of SiO:

$$\lambda = \frac{c}{f} = 2{,}65 \cdot 10^{-7}\ m$$
$$= 265\ nm$$



Ar-H_2 provides most photons of suitable energy

Moreover, H_2 can be dissociated to H and thus induce chemcial cleaning by formation of water.

Moreover, Ar ions can feature high ion energy and induce cleaning via sputtering

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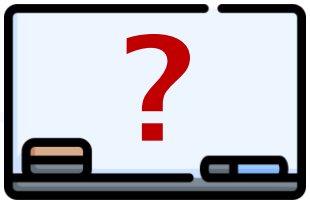
Precision cleaning



- **Worked example 3.3**
  - For the removal of oxide layers from metal surfaces, a cost efficient and effective plasma-based precision cleaning setup should be realised.

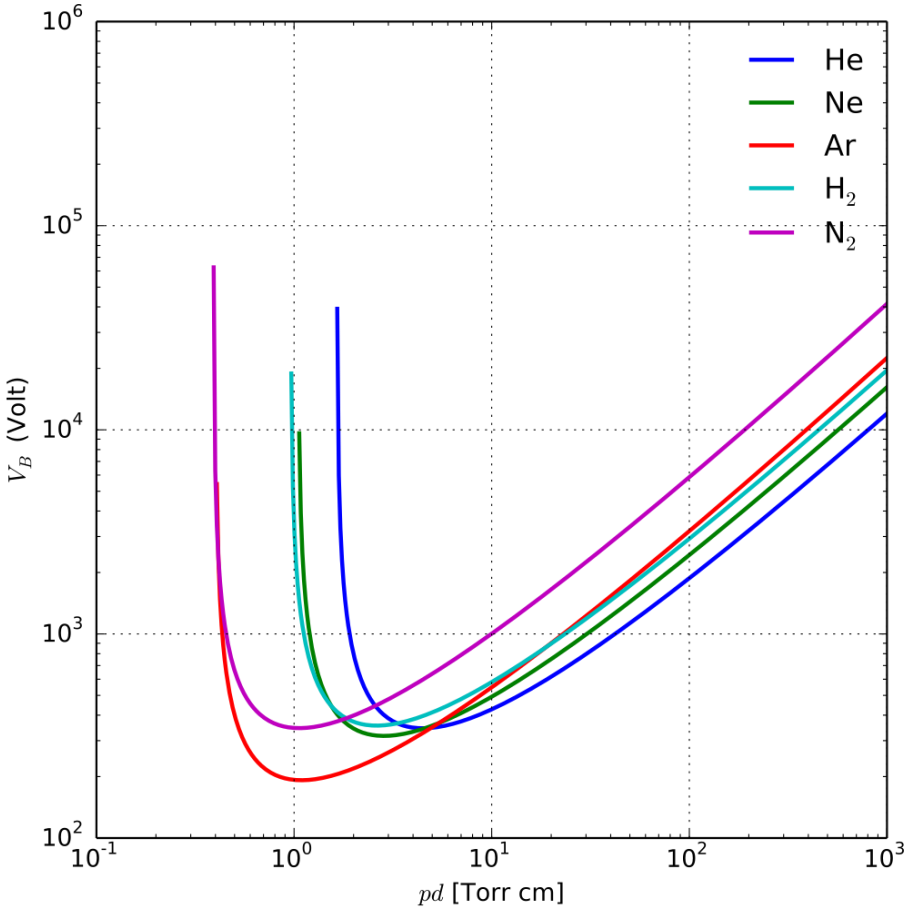
  - Which process gas do you use/choose,
    - forming gas 90/10 or
    - Varigon?

We assume that both gases cost the same price per litre
(which is – to be honest – not true…)

Precision cleaning

- **Worked example 3.3 – Answer**
  - Both available gases feature the same price. In order to realise an efficient plasma process, the required electric energy thus matters.

    Hence, Varigon should be chosen since its main compound, argon, features a notably lower ignition voltage than the main compound of forming gas 90/10, nitrogen, as visualised by the comparison of the Paschen's curve of both gases.



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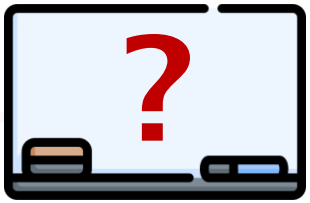
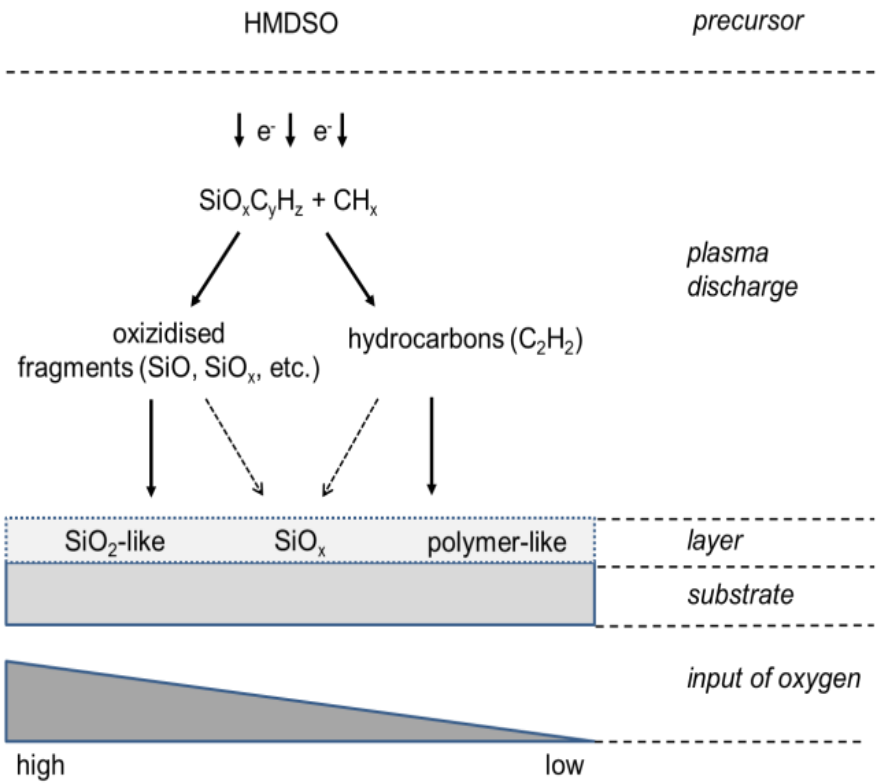
Removal of coatings



- **Worked example 3.4**
  - In order to avoid the reaction of (slightly acidic) lemonade with the plastic body of one-way bottles, the inner surface of such bottles is usually plasma-coated with an anti-diffusion layer. For this purpose, the precursor HMDSO and the following plasma process gases are available:
    - Forming gas 90/10 (90% nitrogen, 10 % hydrogen)
    - Varigon (90% argon, 10 %hydrogen)
    - Air
  - Which gas should be chosen for the above-mentioned application / intended effect?

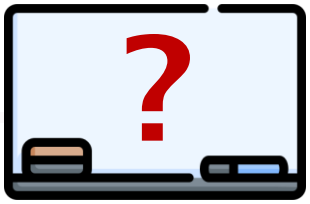
Removal of coatings

- **Worked example 3.4 – Answer**
  - Both forming gas 90/10 and Varigon do not contain any oxygen but 10% of hydrogen, the resulting layer when using these gases is thus most probably polymer-like:



  - In contrast, air provides oxygen, allowing for the growth of a silicon dioxide-like layer. In contrast to polymers, silicon dioxide is – almost – chemically inert. Hence, air should be chosen as process gas…and its for free!

Removal of coatings



- **Worked example 3.5**
- Name/list/discuss the possible effects/mechanisms provided by the below-described plasma source which might contribute to the removal of graffiti from sandstone:

- <u>Parameter of the used plasma source</u>
  - Plasma jet (indirect Plasma)
  - Plasma diameter: 1.3 mm
  - Process gas: air
  - Process gas pressure: 0.3 Mpa
  - Process gas flow rate: 34 l/min
  - Plasma power: 140 W
  - Working distance outlet nozzle-work piece surface: 8 mm
  - Gas temperature at working distance: 90 °C

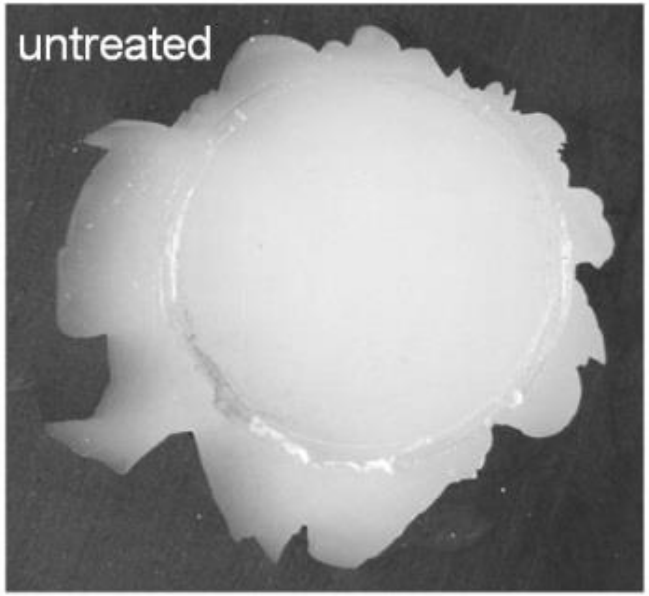
Removal of coatings

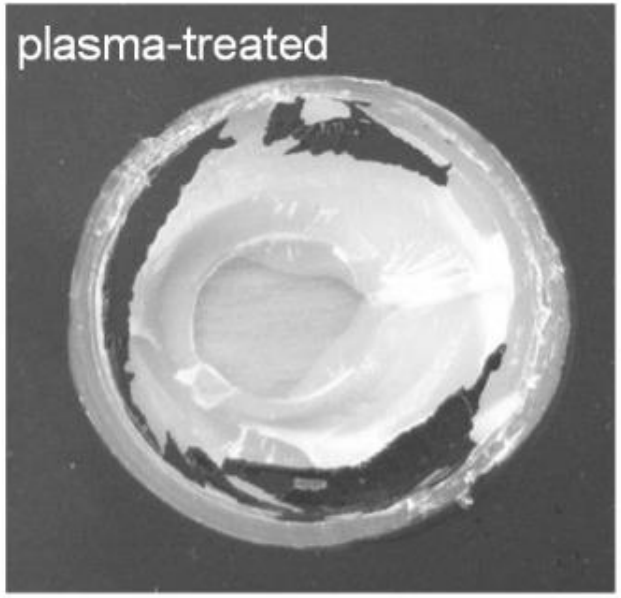
- **Worked example 3.5 – Answer**
- Possible effects:

- <u>Parameter of the used plasma source</u>
  - Plasma jet (indirect Plasma)
    → transport and evacuation of reaction products by process gas flow
  - Plasma diameter: 1.3 mm
    → high density of charge carriers (ion bombardment, electron collisions…)
  - Process gas: air
    → oxidation (O) and/or reduction (N)
  - Process gas pressure: 0.3 Mpa
    → transport and evacuation of reaction products by process gas flow
  - Process gas flow rate: 34 l/min
    → transport and evacuation of reaction products by process gas flow
  - Plasma power: 140 W
    → bleaching of paint via UV irradiation
  - Working distance outlet nozzle-work piece surface: 8 mm
    → composition of plasma cocktail
  - Gas temperature at working distance: 90 °C
    → plasma heating and thermal decomposition/bleaching of paint

Surface functionalisation

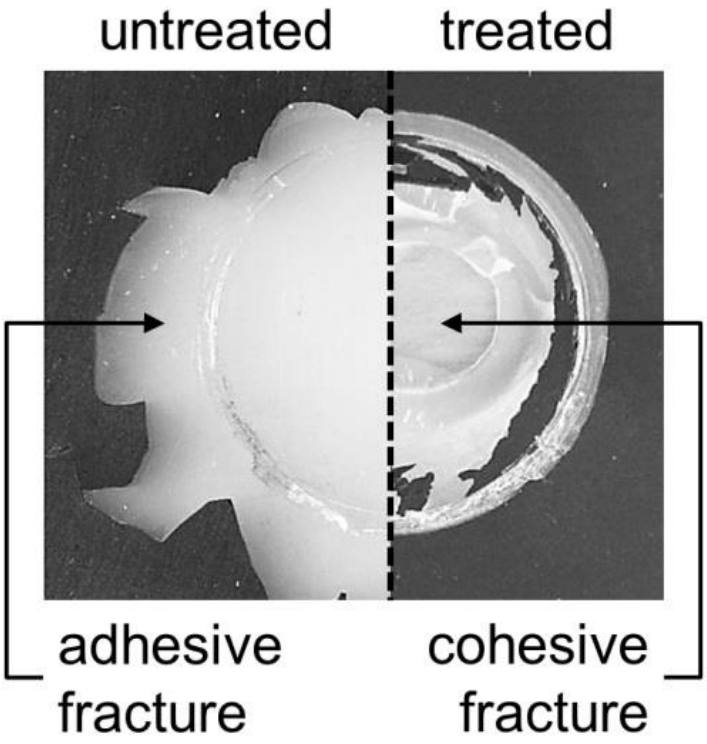
- **Worked example 3.6**
- Discuss the differences in fracture behaviour of paint on untreated and plasma treated on automotive plastics shown below:





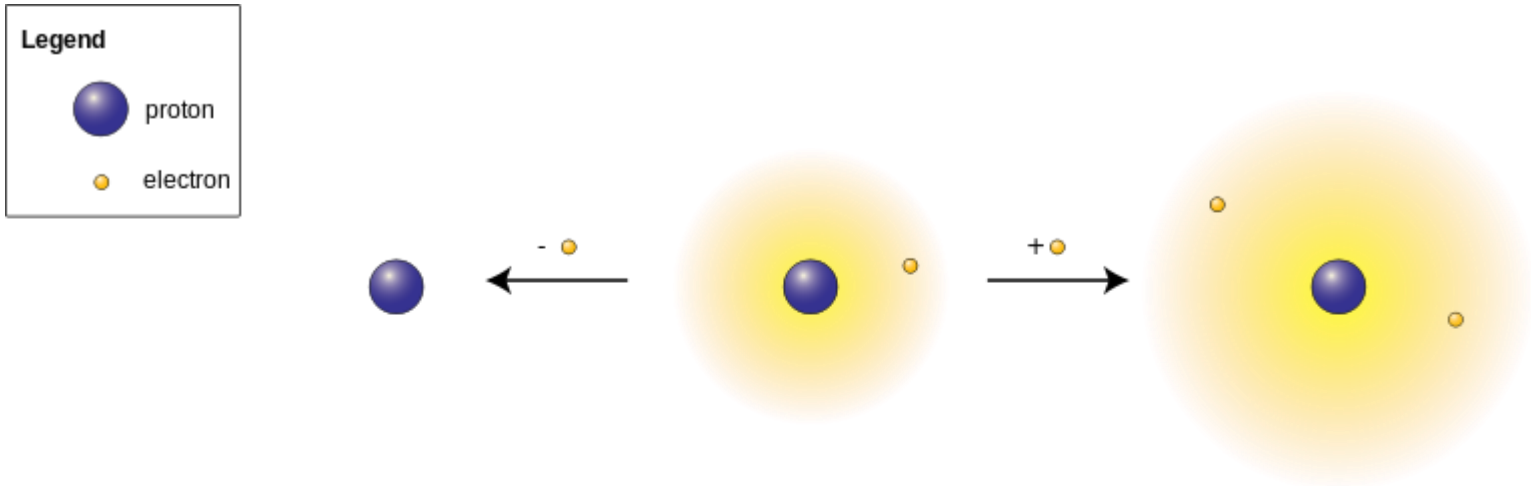
Surface functionalisation

- **Worked example 3.6 – Answer**
  - Untreated: failure between paint and plastic (‘delamination’) = adhesive fracture
  - Treated: internal failure within plastic bulk material = cohesive fracture



Chemical and physical plasma etching ???

- **Exercise 3.1**
- Determine the velocity of a hydrogen ion (H^+) which is accelerated at a voltage of 30 kV.



No. of protons	1	1	1
No. of electrons	0	1	2
Charge	+1	0	-1
Notation	H^+	H	H^-
Classification	cation	neutral (not an ion)	anion

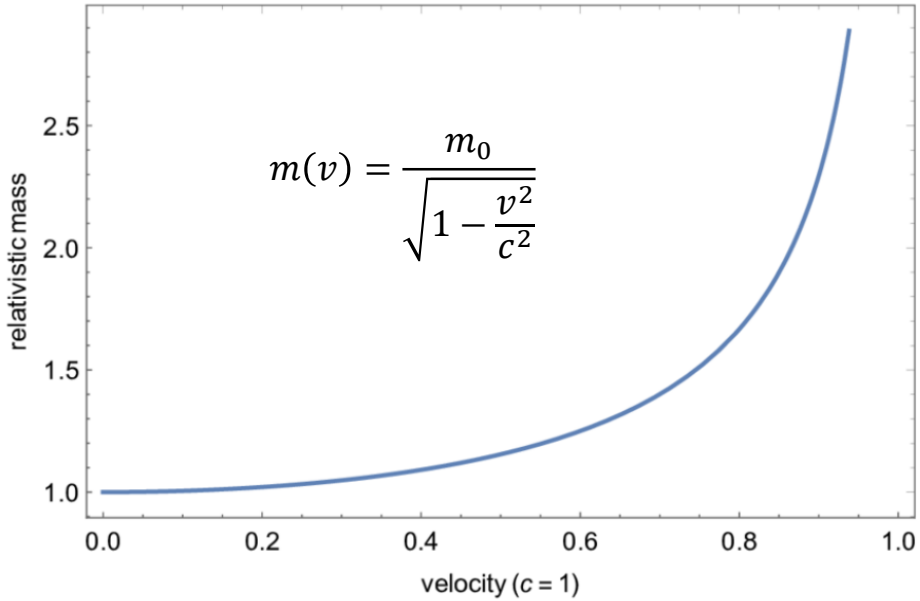
By Jkwchui - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=12617370

Chemical and physical plasma etching

! ! !

- **Exercise 3.1 – Solution**
- The kinetic energy of a hydrogen ion (H^+) accelerated at a voltage of 30 kV amounts to
  $E_{kin} = z \cdot e \cdot U = 1 \cdot 1.6 \cdot 10^{-19} C \cdot 30kV = 4.8 \cdot 10^{-15} J.$
- Its velocity is thus
  $v = \sqrt{\frac{2 \cdot E}{m}} = \sqrt{\frac{2 \cdot 4.8 \cdot 10^{-15} J}{=1{,}67265 \cdot 10^{-27} kg}} \approx 2400000 \;\; m/s.$

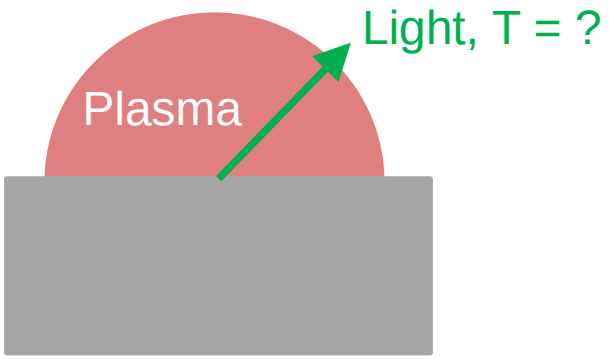
- ≈ 0.8% of the speed of light (so the relativistic mass correction should not be applied to the ion mass:



Plasma effects in laser machining

???

- **Exercise 3.2**
- On a solid surface, a hemispherical plasma with a radius of 22 mm is ignited; its electron density is $10^{15}/m^3$. Determine/estimate the transmission of light emitted from the centre of the plasma volume.



Plasma effects in laser machining

! ! !

- **Exercise 3.2 – Solution**

The plasma frequency is given by

$$\omega_p = \sqrt{\frac{e^2 \cdot n_e}{m_e \cdot \epsilon_0}}$$

and thus amounts to

$$\omega_p = \sqrt{\frac{(1{,}60218 \cdot 10^{-19} As)^2 \cdot \frac{10^{15}}{m^3}}{9{,}10938 \cdot 10^{-31} kg \cdot 8{,}85419 \cdot 10^{-12} \frac{As}{Vm}}} = 1.783.990.256 s^{-1} \approx 1{,}78 GHz$$

where

- e = 1,60218·10^{-19} A·s,
- m_e = 9,10938·10^{-31} kg und
- ε_0 = 8,85419·10^{-12} A·s·V^{-1}·m^{-1}.
- Note: $1V = 1\frac{kg \cdot m^2}{A \cdot s^3}$

Plasma effects in laser machining

! ! !

- **Exercise 3.2 – Solution**

The absorption coefficient of the plasma is then

$$\alpha \approx 2 \cdot \frac{\omega_p}{c} = 2 \cdot \frac{1.78 \cdot 10^9 s^{-1}}{3 \cdot \frac{10^8 m}{s}} = 11.9/m$$

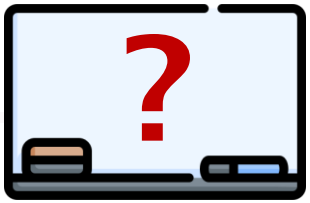
The transmission *T* follows from the Lambert-Beer law and amounts to

$$T = \frac{I_d}{I_0} = e^{-(\alpha \cdot d)} = e^{-(11.9/m \cdot 0.022m)} \approx 0.77 = 77\%$$

The plasma thus features notable absorption which may have an impact on

- Laser materials processing
  → plasma shielding (4.1.1)
- Laser-induced breakdown spectroscopy
  → self absorption of emission lines (4.2.2)

Removal of coatings



- **Worked example 3.7**
- Discuss the consequence of the interrelationship

$$n_{ecrit} = \frac{m_e \cdot \varepsilon_0}{e^2} \cdot f^2$$

Removal of coatings

- **Worked example 3.7 – Answer**

Constant factor,
proportionality constant

$$n_{ecrit} = \boxed{\frac{m_e \cdot \varepsilon_0}{e^2}} \cdot \boxed{f^2}$$

The occurrence of the critical electron density merely depends on the frequency/the wavelength of incident laser irradiation.

Critical electron density is (easily) obtained in the UV wavelength range.